



Tribhuvan University

Faculty of Humanities and Social Sciences

DigiMart: Smart Platform for Electronic Products

A PROJECT REPORT

Submitted to
Department of Computer Application
D.A.V. College

In partial fulfillment of the requirements for the Bachelor of Computer Applications

Submitted by
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November, 2025

Under the Supervision of
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Supervisor's Recommendation

I hereby recommend that this project prepared under my supervision by **HIMAL SUBEDI** entitled “**DIGIMART: SMART PLATFORM FOR ELECTRONIC PRODUCTS**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application be recommended for the final evaluation.

.....

Er. Ishan Pokhrel

SUPERVISOR

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LETTER OF APPROVAL

This is to certify that this project prepared by **HIMAL SUBEDI** entitled “**DIGIMART: SMART PLATFORM FOR ELECTRONIC PRODUCTS**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application has been evaluated. In our opinion, it is satisfactory in the scope and quality of a project for the required degree.

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With Regards

Himal Subedi

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ABSTRACT

DigiMart is a web-based electronics marketplace that includes product recommendation and sales forecasting features. The recommendation system uses content-based filtering to create user preference profiles from viewed products, cart items, and completed orders. Products are ranked using a weighted combination of traditional and description similarity. Traditional similarity is based on product category, brand, price, and rating, while description similarity uses key-value specifications extracted from product descriptions.

For sales forecasting, DigiMart uses a SARIMA model implemented from scratch to predict revenue and product-category demand for a 15-day period. Model selection is performed using chronological rolling-origin validation, with a separate final period reserved for evaluation. The selected model uses two recent non-seasonal values, weekly differencing, and a weekly moving-average term. The forecasts show a recurring seven-day pattern rather than a constant projection. On the final test period, the model achieved a MAPE of 16.97%, compared with 20.28% for the naive baseline.

The system is built using React for the user interface, Express for the application service, and MongoDB for data storage. DigiMart combines online shopping, personalized product recommendations, and sales forecasting in one marketplace system. It supports the customer shopping process while also providing sales information for marketplace management.

Keywords: *DigiMart, Content-Based Filtering, Product Recommendation, SARIMA, Time Series Forecasting, Revenue Forecasting*

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ACF	Autocorrelation Function
ADF	Augmented Dickey-Fuller
API	Application Programming Interface
ARIMA	Autoregressive Integrated Moving Average
CSS	Conditional Sum of Squares
GUI	Graphical User Interface
KPSS	Kwiatkowski-Phillips-Schmidt-Shin
MAE	Mean Absolute Error
MASE	Mean Absolute Scaled Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
NPM	Node Package Manager
PI	Prediction Interval
RMSE	Root Mean Squared Error
RMSSE	Root Mean Squared Scaled Error
SARIMA	Seasonal Autoregressive Integrated Moving Average
VS Code	Visual Studio Code
WAPE	Weighted Absolute Percentage Error

CHAPTER 1: INTRODUCTION

1.1 Introduction

DigiMart is a web-based electronics marketplace that combines online shopping with product recommendation and sales forecasting features. The recommendation system uses content-based filtering to suggest products based on items a user has viewed, ordered, or added to their cart. The forecasting module uses a SARIMA model to estimate future revenue and product demand from historical sales data. Customers can browse products, place orders, and receive personalized recommendations, while administrators can manage products, users, and orders and view sales forecasts and related information. By bringing these features together, DigiMart provides a marketplace platform that supports both the customer shopping experience and the management of sales and inventory.

1.2 Problem Statement

Online electronics marketplaces offer a large number of products, making it difficult for customers to find products that match their interests and previous shopping activity. Without personalized recommendations, customers may need to manually search through many products before finding suitable options. At the same time, marketplace administrators need to monitor sales and demand in order to make informed decisions about products and stock levels. Relying only on historical sales data makes it difficult to estimate future demand and revenue, particularly when sales patterns change over time.

DigiMart addresses these problems by combining a content-based product recommendation system with a time series forecasting module within an online electronics marketplace. The recommendation system uses information from user interactions, including viewed products, cart items, and completed orders, to identify products with similar characteristics. The forecasting module uses historical sales data to estimate future revenue and product category demand. The project therefore aims to provide customers with more relevant product recommendations while giving administrators useful information for monitoring sales and planning stock.

1.3 Objectives

The main objective of this project is:

- To develop a web-based DigiMart application for buying and selling electronic products, including the management of products, orders, stock and users.
- To implement a content-based product recommendation system and a SARIMA-based forecasting system for providing personalized product recommendations and estimating future revenue and product demand.

1.4 Scope and Limitations

DigiMart covers the development of a web-based electronics marketplace where customers can browse products, view product details, add items to their cart, place orders, and receive personalized product recommendations. The system also provides administrators with features to manage products, users, orders, and stock. In addition, the project includes a content-based recommendation system based on user activity and product attributes, along with a SARIMA based forecasting system for estimating future revenue and product category demand from historical sales data.

Limitations

- The recommendation system uses only content-based filtering and does not consider similarities between different users or use collaborative filtering.
- The forecasting results depend on the available historical sales data and may be affected by changes in customer behavior, product availability, promotions, and other external factors that are not included in the model.

1.5 Development Methodology

The Iterative Waterfall model was chosen because DigiMart was developed in defined phases. Requirements, design, implementation, testing, and deployment were completed in sequence, while feedback allowed earlier work to be revised when necessary. This process supported clear milestones, regular documentation, and gradual integration of the recommendation and forecasting modules.

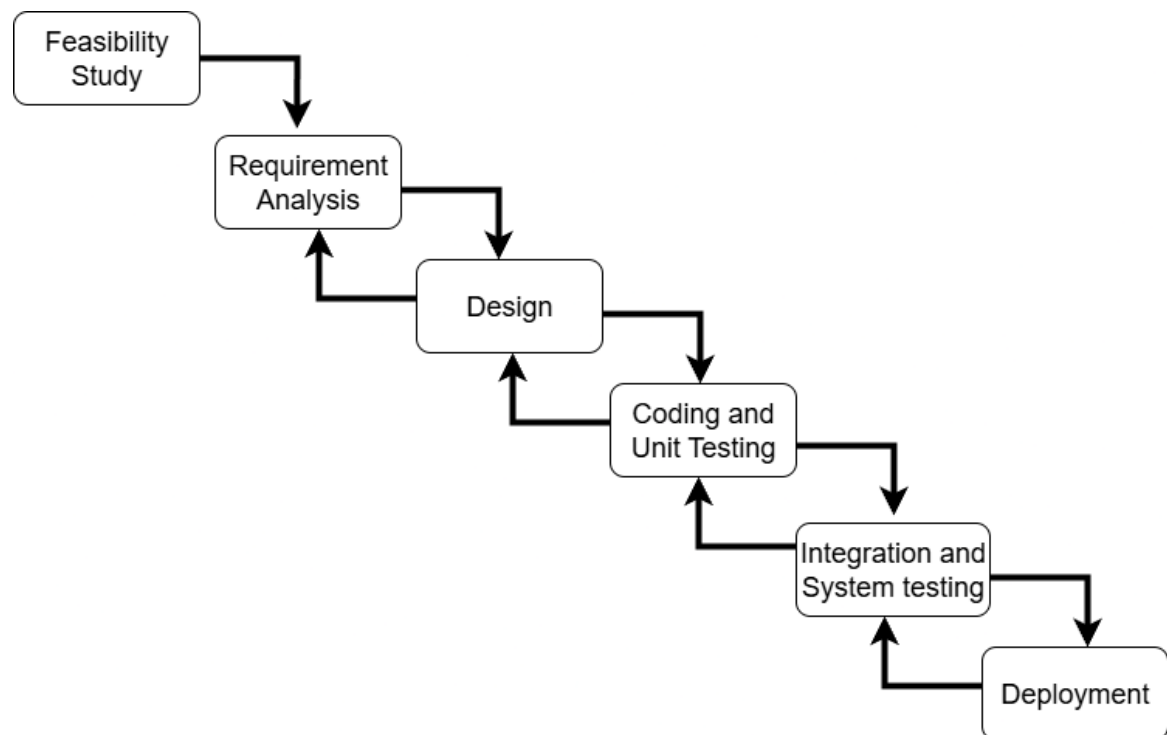


Figure 1.1: Iterative Waterfall Model

1.6 Report Organization

- **Chapter 1: Introduction:** Provides an overview of the DigiMart project, including its background, motivation, problem statement, objectives, scope and limitations, and development methodology.
- **Chapter 2: Background Study and Literature Review:** Reviews concepts related to product recommendation, sales forecasting, and the technologies used in the project, including content-based filtering, SARIMA forecasting, and relevant previous studies.
- **Chapter 3: System Analysis and Design:** Describes the functional and non-functional requirements, feasibility analysis, system architecture, database design, and diagrams used to model DigiMart.
- **Chapter 4: System Implementation and Testing:** Describes the tools, programming languages, libraries, and frameworks used to develop DigiMart. It presents the implementation of the content-based recommendation and SARIMA forecasting modules, their integration with the web application, and the testing performed on the system.
- **Chapter 5: Conclusion and Future Work:** Summarizes the project results, limitations, and possible future work.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

The DigiMart platform addresses the needs of both customers and business administrators in an electronics marketplace. Customers need convenient product browsing and relevant product suggestions, while administrators need tools to manage products, orders, stock, and sales information. These requirements provide the basis for including product recommendation and sales forecasting features in the system.

Product recommendation systems help customers find relevant products from a large collection of available items. Content-based filtering is suitable for DigiMart because recommendations can be generated from product attributes and the customer's previous interactions with products. Sales forecasting is also useful for marketplace management because historical sales data can be used to estimate future revenue and product demand. Time series models such as SARIMA can be used to model patterns in historical sales data, including recurring weekly patterns.

DigiMart implements content-based product scoring and a SARIMA forecasting framework from scratch. The forecasting models are evaluated using chronological forecast origins, with a separate final period reserved for testing. Simple baseline methods and a reference implementation are also used for comparison. This provides a basis for evaluating the forecasting results and the custom implementation used in DigiMart.

2.2 Literature Review

Basu, Hirsh, and Cohen studied recommendation as a classification problem using both content information and user preference data [1]. Their method represented items using descriptive attributes and combined these attributes with rating information to predict whether a user would prefer an item. The evaluation used more than 45,000 movie ratings and showed that including content information improved precision compared with methods based only on collaborative data. This study is relevant to DigiMart because it demonstrates the usefulness of item attributes for generating recommendations. DigiMart applies a similar content-based approach by using product category, brand, price, rating, and description features together with information from viewed products, cart items, and previous orders. However, DigiMart uses a content-based method rather than the hybrid classification approach proposed in the study.

Ramos, Santos, and Rebelo compared state space models and ARIMA models for sales forecasting at Foreva, a Portuguese footwear retailer [2]. Their data covered five footwear categories with different demand patterns. Forecast performance was evaluated using RMSE, MAE, and MAPE for both one-step and multi-step forecasts. Neither approach consistently performed better across all categories, indicating that forecasting performance depends on the characteristics of each time series. The study also distinguished between model fit and performance on observations outside the training data. These findings are relevant to DigiMart because its forecasting module considers both total revenue and product category quantities. They also support the use of chronological evaluation rather than relying only on in-sample model fit.

Padilha, Pereira, and Albarracin compared SARIMA and Neural Network AutoRegressive models for short-term sales forecasting in fashion e-commerce [3]. Most of the daily sales series showed a seven-day pattern, leading the authors to evaluate models with a weekly seasonal period, represented as $SARIMA(p, d, q)(P, D, Q)_7$. Forecasts were evaluated on the final week of each sales stage using RMSE, SMAPE, and TAPE. Neural network models generally performed better, while SARIMA remained competitive for more stable periods. This study is relevant to DigiMart because it supports the use of a weekly seasonal period for daily retail sales data and shows the importance of evaluating model performance on different types of demand patterns.

Hyndman and Athanasopoulos describe rolling-origin evaluation as a time-series cross-validation method in which each forecast is generated using only observations available at the corresponding forecast origin [4]. Hyndman and Koehler propose MASE as a scale-free error measure that allows forecasting performance to be compared across different series and against a naive method [5]. These approaches are relevant to DigiMart because the forecasting model is selected using chronological forecast origins and evaluated on a separate final test period. ADF and KPSS tests provide statistical methods for assessing stationarity and supporting differencing decisions [6, 7], while the Ljung-Box test is used to examine residual autocorrelation [8]. Together, these studies provide the theoretical basis for the forecasting model selection, evaluation, and diagnostic procedures used in DigiMart.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

System analysis identifies the requirements and functions of the DigiMart system. It covers the main marketplace activities, including product browsing, cart and order management, user and product management, stock management, product recommendation, and sales forecasting. It also considers non-functional requirements such as usability, security, performance, and maintainability.

3.1.1 Requirement Analysis

For the DigiMart application, the requirements are classified into functional and non-functional requirements as follows:

i. Functional Requirements:

Functional requirements describe the main functions and services that the system must provide. The key functional requirements include:

- **Authentication and Authorization:** Allow users to register and log in, and restrict administrative functions to authorized accounts.
- **Product Catalogue:** Allow users to search, browse, and view product details.
- **Cart and Orders:** Allow users to manage cart items, provide shipping information, place orders, and view order history.
- **Reviews:** Allow authenticated users to submit product ratings and comments.
- **Product Recommendation:** Rank products using content-based scoring based on viewed products, completed orders, and cart contents.
- **Sales Forecasting:** Display revenue and product-category forecasts along with evaluation metrics, baseline comparisons, model diagnostics, and prediction intervals.
- **Administration:** Allow administrators to manage products and users, manage stock, and review customer orders.

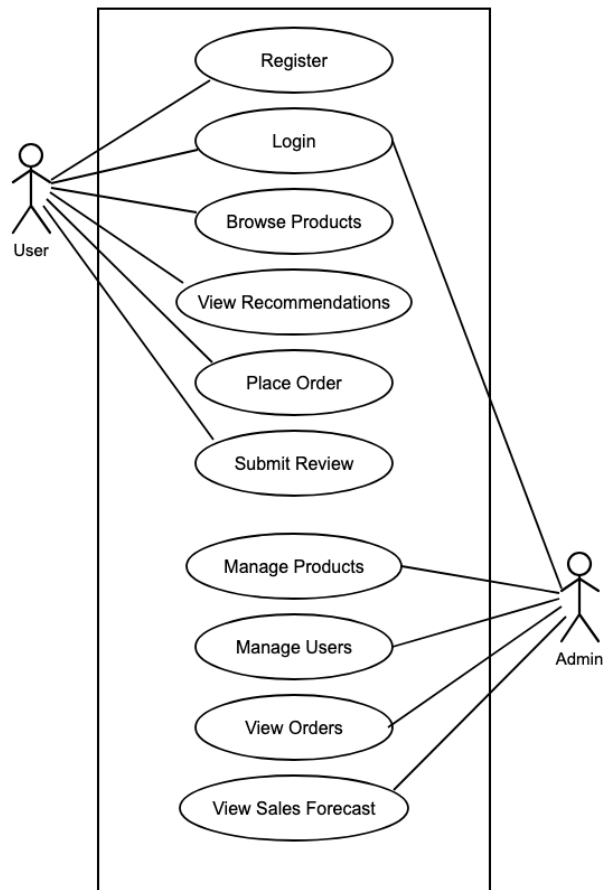


Figure 3.1: Use Case Diagram of DigiMart

ii. Non-Functional Requirements:

Non-functional requirements describe the quality and operational requirements of the system. These include:

- **Usability:** The interface should be clear and usable on both desktop and mobile screens.
- **Security:** Passwords should be hashed, authenticated requests should use signed tokens, and administrative routes should enforce role-based access checks.
- **Reliability:** Invalid input and unavailable services should return controlled errors instead of incomplete or misleading responses.
- **Performance:** Pages and API responses should load within a reasonable time under normal project usage.
- **Maintainability:** Marketplace, recommendation, forecasting, and interface code should be organized into identifiable modules.

3.2 Feasibility Analysis

1. Technical Feasibility

The DigiMart platform is technically feasible because it uses established open-source technologies. React is used for the user interface, Node.js and Express for the application service and recommendation logic, MongoDB for data storage, and Python for the forecasting module. The required tools and libraries are freely available. Forecasting results are evaluated using a separate test period.

2. Operational Feasibility

DigiMart provides customers with a searchable product catalogue, cart and order functions, and product recommendations. Administrators can manage products, users, orders, and stock, and view sales forecasts and evaluation information. These functions support the intended marketplace activities.

3. Economic Feasibility

The DigiMart project uses open-source technologies and does not require software licensing fees. The required development tools, frameworks, and libraries are freely available, making the project suitable for development within the available resources.

4. Scheduling Feasibility

The project is feasible within the planned timeline. Each phase is organized into defined tasks, milestones, and deadlines. The Gantt chart below presents the planned schedule and distribution of project activities.

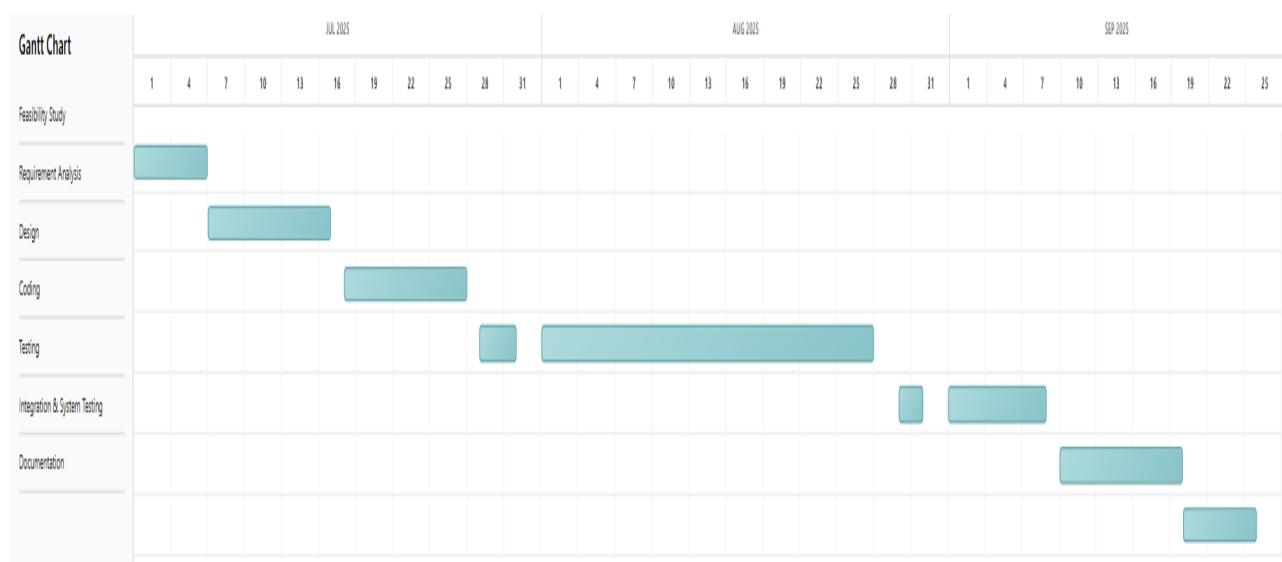


Figure 3.2: Gantt Chart

3.3 Object Modelling Using Class Diagram

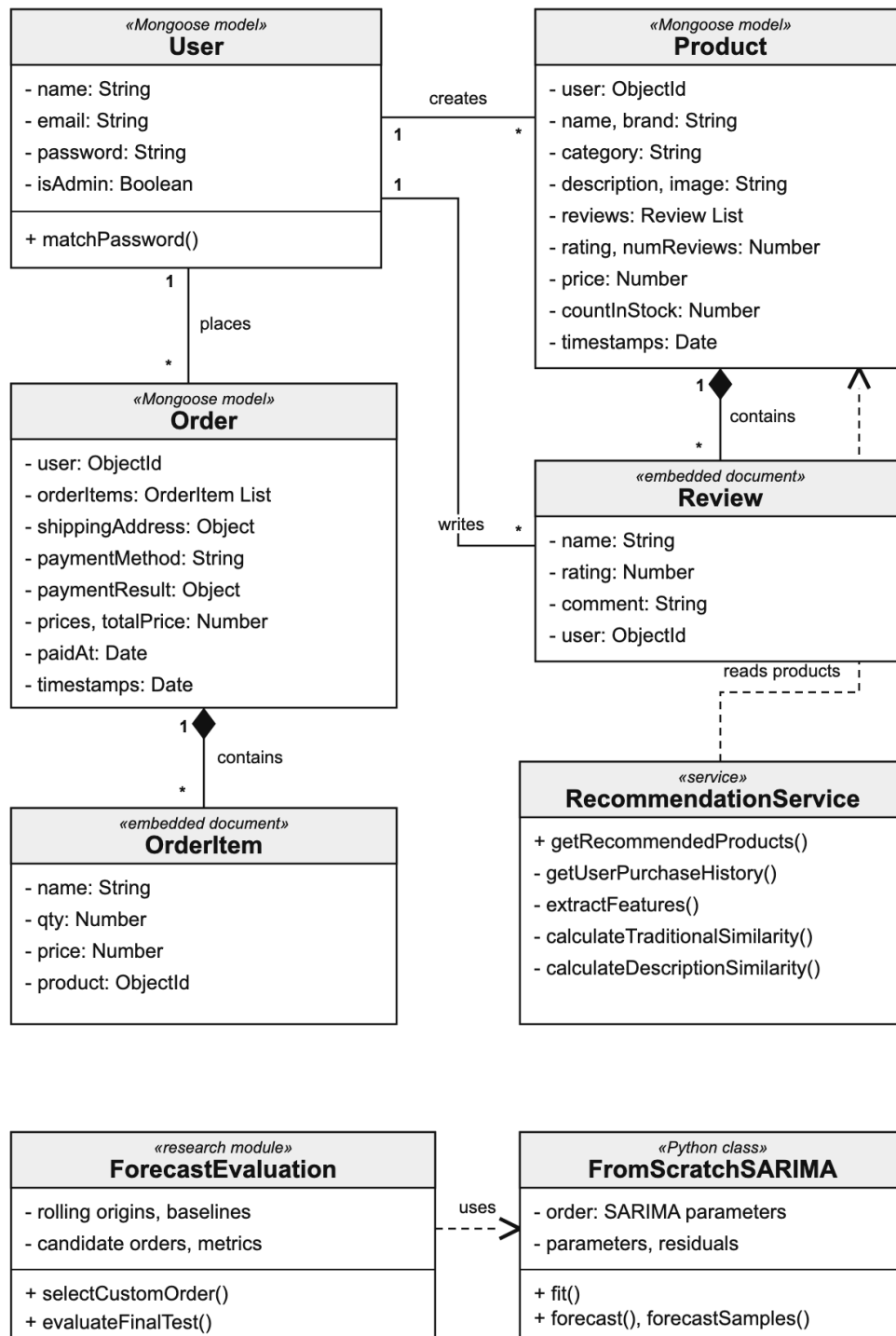


Figure 3.3: Class Diagram of DigiMart

The DigiMart recommendation service reads product views, cart items, and completed orders when a recommendation request is made. It builds a user preference profile and applies deterministic similarity rules; it does not continuously train a recommendation model. The forecasting research module is independent of MongoDB. It evaluates the prepared sales dataset and writes versioned JSON artifacts used by the application.

3.3.1 Dynamic Modelling Using Sequence Diagram

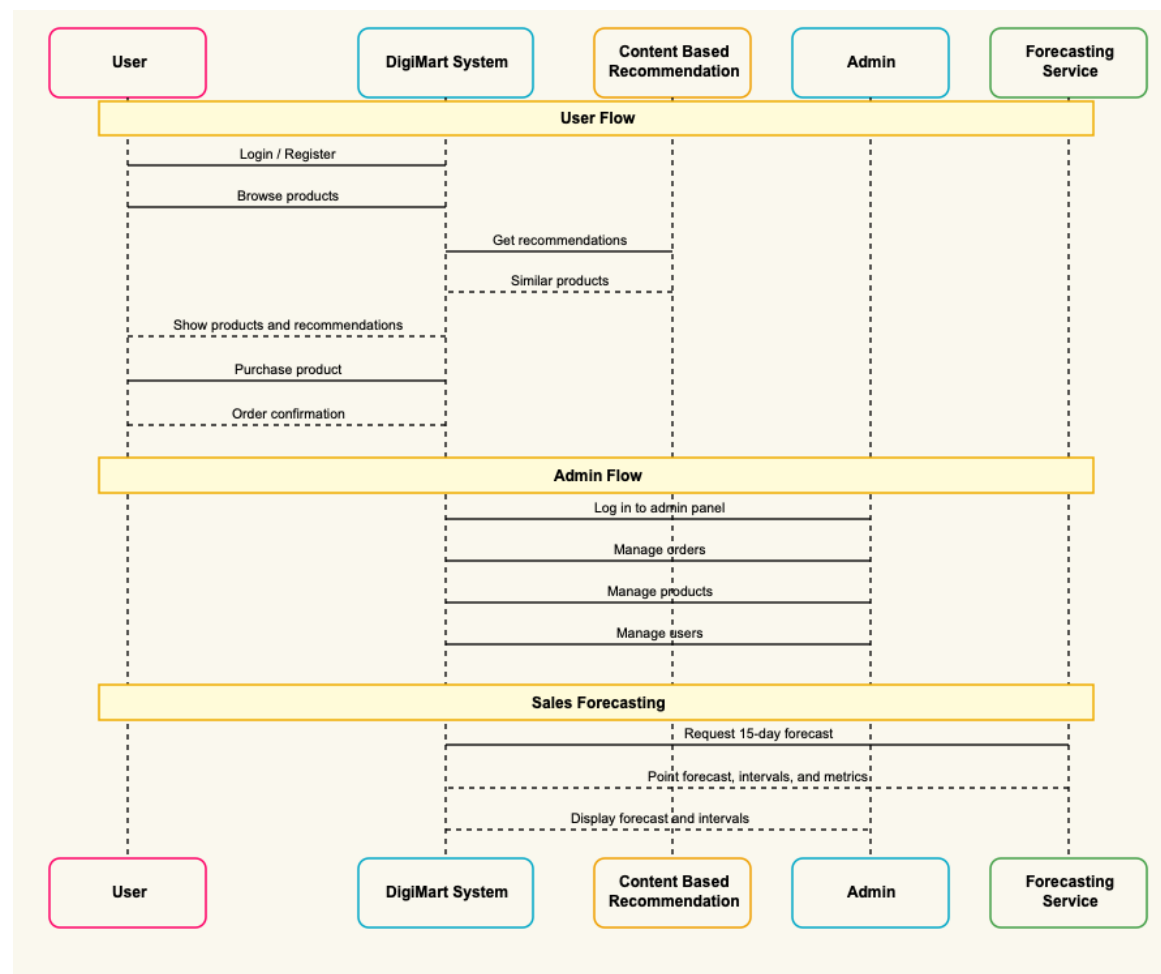


Figure 3.4: Sequence Diagram of DigiMart

Users register or log in, browse products, receive ranked recommendations, and place orders. Administrators manage products, users, and orders. Opening the forecasting dashboard causes Express to read the saved 15-day forecast and metrics artifacts directly. These artifacts provide the point forecasts, 80% prediction interval, and evaluation results without fitting the model. An administrator can use the Recalculate Forecast action to start the forecasting service and generate updated artifacts when required.

3.3.2 Process Modelling Using Activity Diagram

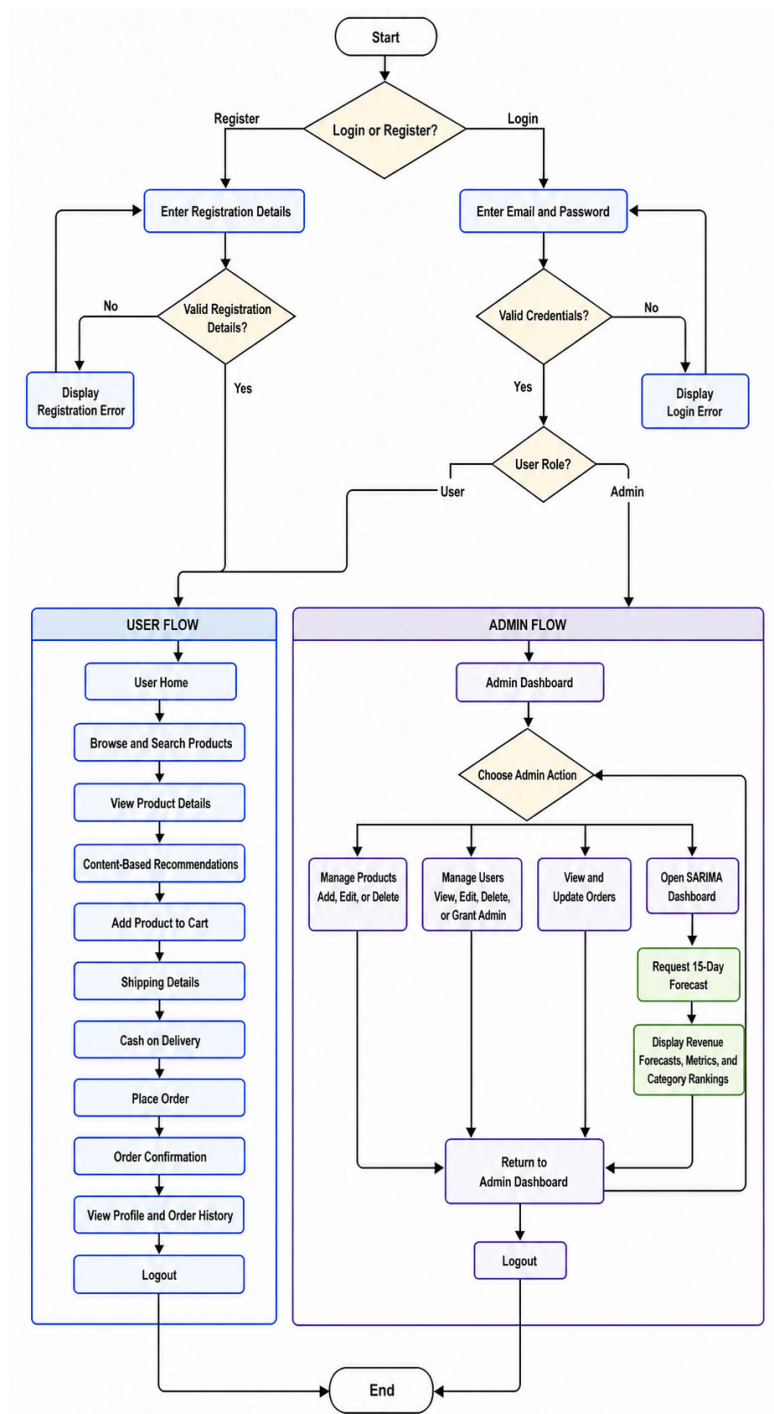


Figure 3.5: Activity Diagram of DigiMart

The activity diagram shows the main user and administrator operations. After authentication, an administrator can manage products and users, view orders, or open the forecasting dashboard. A customer can browse products, view details and recommendations, add products to the cart, select Cash on Delivery, place an order, and receive confirmation. Decision points cover invalid credentials, continued browsing, and the login requirement at checkout.

3.3.3 Database Schema Diagram

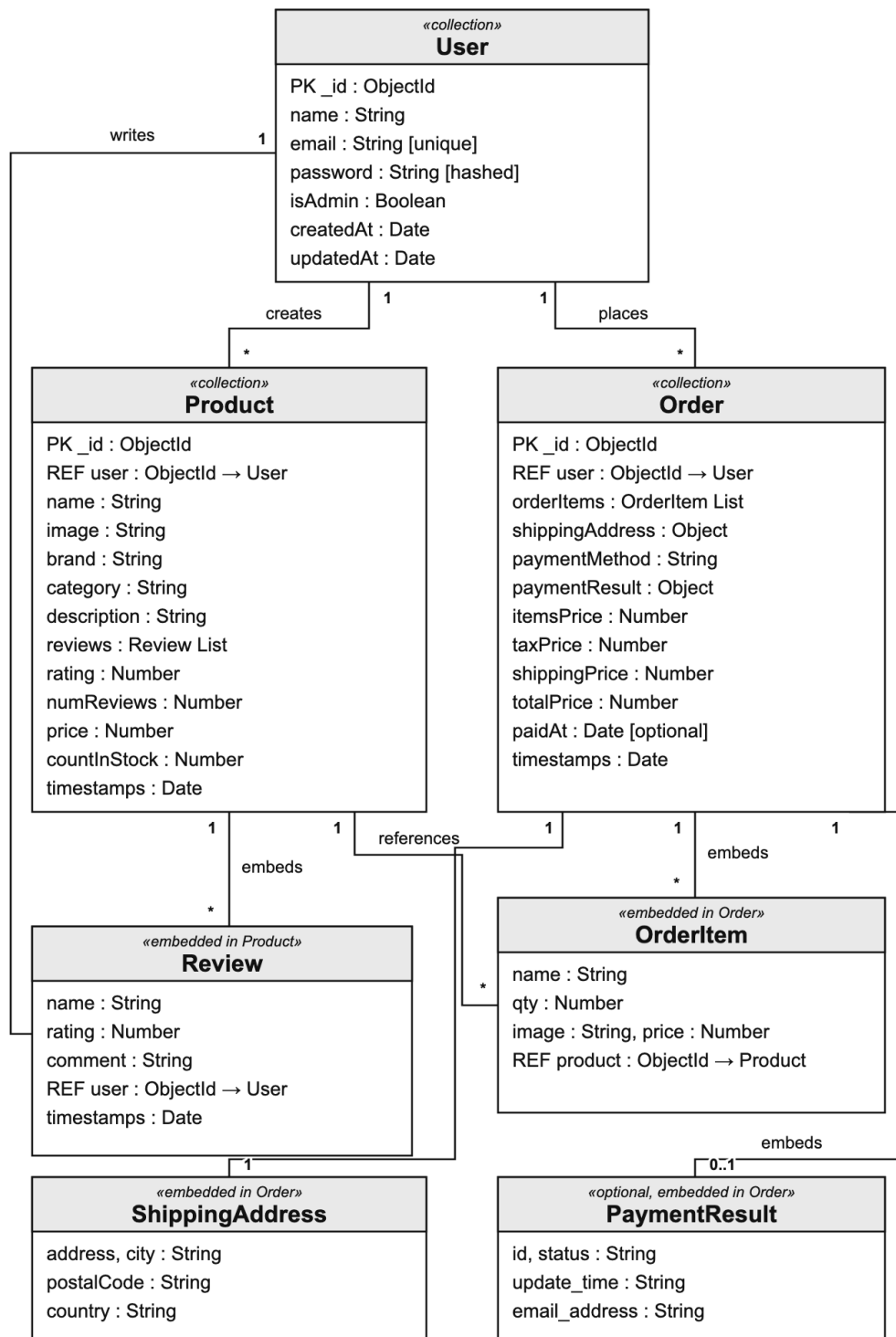


Figure 3.6: Database Schema Diagram of DigiMart

This database schema represents the marketplace data stored by DigiMart. A User can create Products, place Orders, and write Reviews. A Product embeds its Reviews, while an Order embeds OrderItems and a shipping-address object. Each OrderItem references one Product. Payment information is stored as an object within the Order document. The separate forecasting CSV and generated forecast artifacts are not MongoDB entities and are therefore not shown in this schema.

3.3.4 Algorithm Details

Content-Based Recommendation: The content-based recommendation system suggests products by comparing stored attributes and colon-delimited key-value specifications parsed from product descriptions with a user preference profile:

1. **Check User Activity:** Determine the source of recommendation request.
 - **No Activity:** Return the latest products sorted by creation date in Explore Mode and terminate the process.
 - **Has Activity:** Use cart items, viewed products, and purchase history to build the user profile.
2. **Build User Profile:**
 - Aggregate the last 10 products from each relevant source based on activity type.
 - Extract key value features from product descriptions using pattern matching.
 - Aggregate features across all reference items, collecting multiple values for each attribute.
 - Calculate average price and average rating from valid items.
 - Compile unique categories and brands from reference products.
3. **Get All Products:** Retrieve all products from the database for scoring.
4. **Score Each Product:** For every candidate product, calculate two similarity components:
 - (a) **Traditional Similarity with 40% weight:**
 - Category Match: 40% if product category matches profile categories
 - Brand Match: 30% if product brand matches profile brands
 - Price Similarity: 20% based on normalized difference from profile average price
 - Rating Similarity: 10% based on normalized difference from profile average rating

(b) **Description Similarity with 60% weight:**

- Extract key value features from the product description
- Compare each product feature against profile features
- Assign match scores: 1.0 for exact match, 0.5 for partial match, 0 for no match
- Calculate average match score across all comparable features
- Track matching attributes with their product and profile values

(c) **Compute Final Score:** Calculate weighted combination of traditional and description similarities.

(d) **Add Metadata:** Flag products that are currently in cart or previously purchased.

5. **Sort and Paginate:**

- Sort all products by final similarity score in descending order.
- Assign ranks to products based on similarity.
- Apply pagination: full page for general recommendations, top 4 for product page.

6. **Return Results:** Output includes ranked products with similarity scores, detailed scoring breakdown, feature match details, user profile summary, and product status flags.

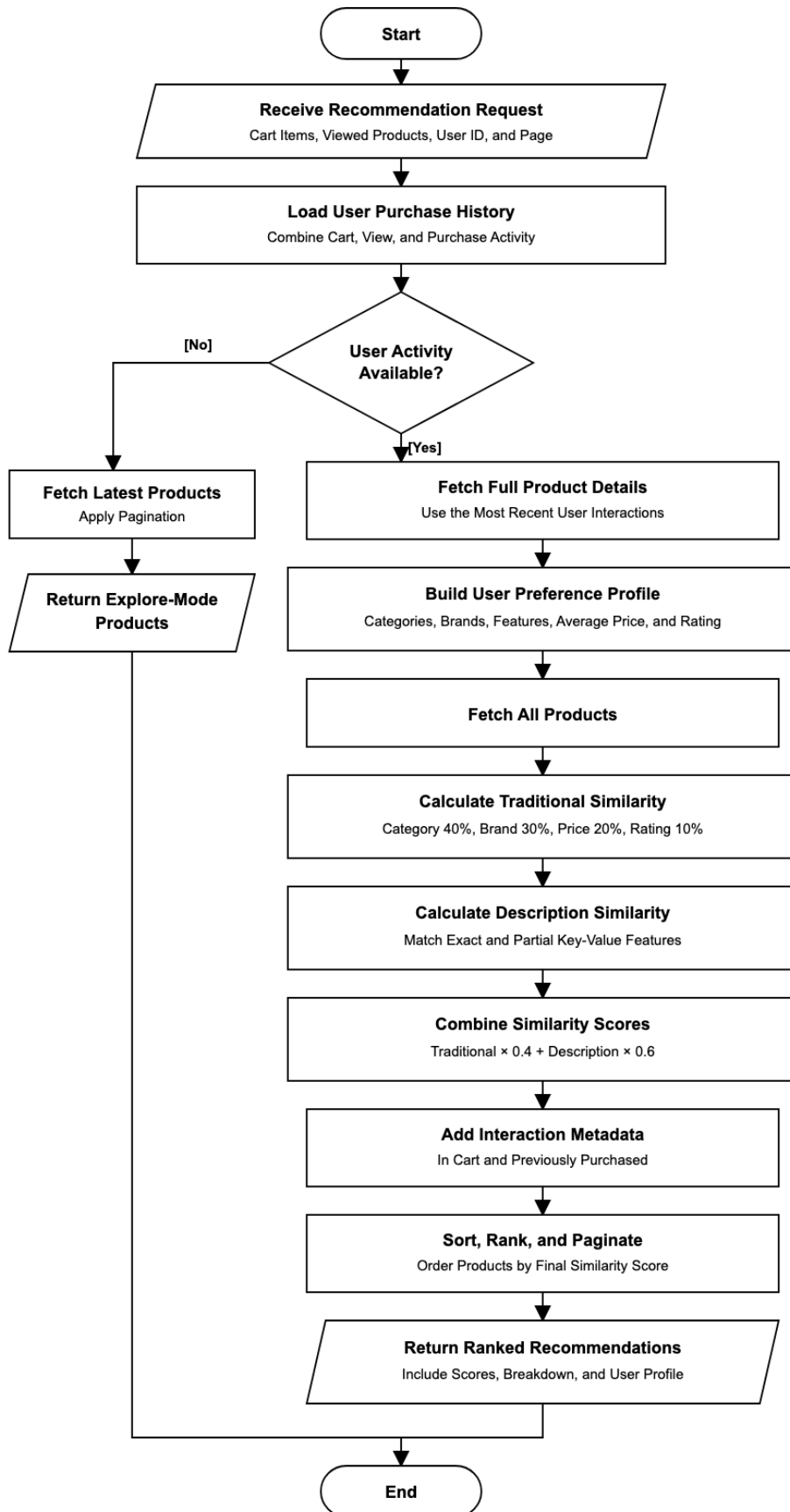


Figure 3.7: Content-Based Recommendation Flowchart

Time Series Forecasting with SARIMA: The forecasting module implements a multiplicative SARIMA framework from scratch. In $\text{SARIMA}(p,d,q)(P,D,Q)_s$, p,d,q denote the non-seasonal orders, P,D,Q denote the seasonal orders, and $s = 7$ represents the weekly seasonal period. NumPy is used for log transformation, differencing, multiplicative AR and MA recursion, conditional residual calculation, and Gauss-Newton estimation using Conditional Sum of Squares (CSS). Scikit-learn is used for standard error calculations, SciPy for chi-square probabilities in residual testing, and statsmodels for stationarity tests and independent ARIMA and SARIMA reference implementations [9].

1. Prepare the Primary Series:

- Load completed transactions from the cleaned customer data CSV file. Aggregate revenue by date, insert missing calendar dates with zero revenue, and sort the observations chronologically.
- Use all 366 raw daily revenue observations as the primary target. A trailing 3-day mean is evaluated only as an ablation because smoothing changes the target and may reduce apparent forecast error.
- Apply the $\log(1 + y)$ transformation within each fitted revenue model. Product-category quantities are modeled separately using their raw values.

2. Reserve the Final Test Period:

- Use the first 293 observations (80%) for model development and reserve the final 73 observations (20%) for final testing.
- Preserve the chronological order of the series. No final-test observation is used when selecting differencing orders, model orders, or baseline orders.
- Apply ADF and KPSS tests to transformations of the development log series as supporting evidence for stationarity and differencing decisions [6, 7].

3. Construct Horizon-Specific Validation Folds:

- Construct seven expanding rolling origins within the development period.
- Use 188, 203, 218, 233, 248, 263, and 278 available observations at the respective origins, and forecast the following 15 days at each origin.
- At each origin, evaluate training histories of the most recent 91, 126, and 182 days, as well as the complete available history. Common MASE and RMSSE scales are calculated using the complete available history.

4. Select the Custom SARIMA Order:

- Evaluate 72 SARIMA orders under four training-history policies, resulting in 288 configurations evaluated at each validation origin.
- Fit each converged configuration and require stable AR and MA roots and Ljung-Box p -values of at least 0.05 at lags 7, 14, and 21.
- For the deployed revenue forecast, retain adequate candidates containing a seasonal AR, seasonal difference, or seasonal MA term to ensure that weekly seasonality is represented over the 15-day forecast horizon. If no such candidate is adequate, use the unrestricted adequate candidate pool.
- Rank eligible candidates using mean MASE across the seven validation origins. Pooled RMSE and then the number of fitted terms are used as tie-breakers. The unrestricted ranking is also retained in the generated metrics for comparison.
- Use MASE as the primary metric because it is scale-free and compares forecast error against a weekly seasonal-naive scale [5]. RMSSE provides the corresponding squared-error measure, WAPE divides total absolute error by total actual revenue, and MAPE excludes observations with zero actual values.

5. Evaluate Reference Methods:

- Evaluate naive, weekly seasonal-naive, and selected non-seasonal ARIMA baselines using the same validation origins.
- Fit the selected SARIMA order using statsmodels as an independent implementation reference.
- Compare the coefficients obtained from the custom CSS implementation with estimates from the same-order statsmodels model. Differences are expected because CSS and maximum likelihood use different fitting objectives.
- Keep custom-order selection separate from model-family comparison and do not claim that the custom SARIMA model is universally superior to the reference or baseline methods.

6. Perform Rolling Final Testing:

- Refit the selected order using the selected 126-day history at each final-test origin and generate a 15-day forecast.
- Add previously forecast final-test observations to the available history only after their actual values have been observed.
- Cover all 73 final-test observations using five rolling origins.
- Report weighted-mean MAE, MAPE, WAPE, MASE, and RMSSE across the rolling origins. Calculate pooled RMSE from all squared errors rather than averaging the RMSE values of individual folds.

7. Assess Residuals and Forecast Uncertainty:

- For the selected 126-day development fit, examine the residual ACF, Ljung-Box statistics at lags 7, 14, and 21, and the moduli of the AR and MA roots [8].
- Generate 2,000 centered residual-bootstrap paths to construct 80% prediction intervals, and evaluate their coverage on the rolling final-test forecasts.
- Generate the deployed point forecast using zero future innovations and direct inverse log transformation. Prediction intervals are obtained from bootstrap-path quantiles, so the point forecast is not required to equal the interval midpoint.
- Evaluate the bootstrap-path mean separately. Since it performs worse than the conditional point forecast on the development metrics, the conditional point forecast is used for deployment.

8. Generate and Serve Forecast Artifacts:

- Fit the selected revenue model using the most recent 126 days and fit each category model using its complete available series. Store versioned forecast, metric, and metadata information as JSON artifacts.
- Record dataset and pipeline hashes in the generated artifact metadata.
- Store the forecast and metrics as saved JSON artifacts. Express reads these artifacts directly for normal dashboard requests without refitting the models. The protected *Recalculate Forecast* action executes the complete forecasting pipeline and generates updated artifacts when explicitly requested by an administrator.

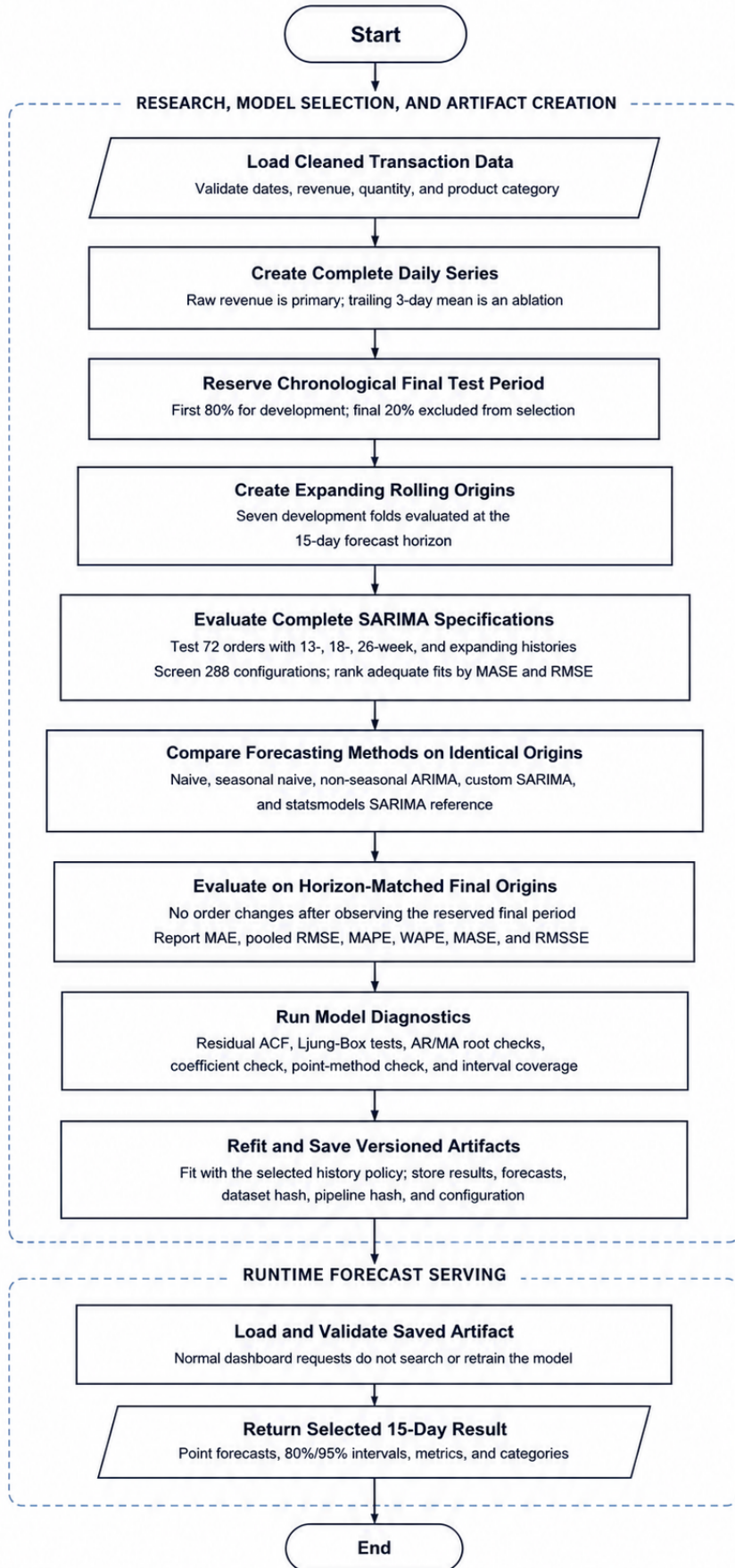


Figure 3.8: SARIMA Forecasting Flowchart

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Development Tools and Technologies

The implementation uses the following technologies and supporting tools:

- **React and JavaScript**

React implements the component-based user interface, while JavaScript is used throughout the frontend and application backend.

- **Node.js and Express**

Node.js runs the application backend, and Express defines marketplace, authentication, recommendation, and forecasting routes.

- **MongoDB and Mongoose**

MongoDB stores marketplace documents, and Mongoose defines schemas, relationships, validation, and database queries.

- **Python and Flask**

Python implements the forecasting research pipeline, while Flask handles forecasting recalculation requests and related service operations.

- **Scientific Python Libraries**

NumPy implements numerical operations, SciPy and scikit-learn support diagnostics and metrics, and statsmodels provides independent reference models and stationarity tests.

- **VS Code, Postman, and MongoDB Compass**

These tools support source editing, API inspection, and database inspection.

- **Git and NPM**

Git records source history, while NPM manages JavaScript dependencies and project scripts.

4.1.2 Implementation Details of Modules

The DigiMart application is divided into modules based on their main responsibilities.

- i. **Recommendation Module** The recommendation module implements content-based product scoring through the `getRecommendedProducts` method. It uses products from the user's browsing history, completed orders, and cart items to create a preference profile. The profile considers product category, brand, price, rating, and key-value specifications extracted from product descriptions. Products are then ranked using a weighted similarity score.
- ii. **Forecasting Module** The forecasting module generates revenue and product-category forecasts using the SARIMA-based forecasting pipeline. The `generate_artifacts.py` pipeline prepares the daily revenue series, reserves the final 20% for testing, and performs model selection using seven rolling origins for the 15-day forecast horizon. It evaluates custom SARIMA candidates, baseline methods, a statsmodels reference implementation, residual diagnostics, and bootstrap prediction intervals. The selected model and evaluation results are stored as JSON artifacts together with data and source-code hashes. Administrators can initiate the forecasting process through the protected *Recalculate Forecast* button. For normal dashboard requests, Express reads the saved forecast and metrics artifacts directly. The forecasting service is used when an administrator initiates forecast recalculation.
- iii. **Authentication** The authentication module handles user registration, login, profile management, and access control. It manages user information and administrator privileges and provides authorized user information to other modules.
- iv. **Database** The database module manages users, products, orders, reviews, order items, shipping addresses, and payment information in MongoDB. Mongoose schemas define the structure, validation, and relationships of the stored documents.
- v. **Frontend** The frontend module provides the user interface for the DigiMart platform. It includes responsive views for product browsing, product details, cart management, order placement, authentication, recommendations, reviews, profiles, view history, and order status. It also displays forecasting results and related charts for administrators.

4.2 Testing

Software testing was conducted to verify individual modules, user-facing functions, and communication between system services. The following tables present the final test results after identified defects were corrected.

4.2.1 Functional Testing

The following cases were verified through the application interface.

Table 4.1: Login and Registration Testing

S.N.	Test Case	Input	Expected Result	Actual Result	Status
1	Login	Valid email and password	Login successful	Login successful	Pass
2	Login	Invalid email or password	Login rejected	Login rejected	Pass
3	Registration	Valid user details	Registration successful	Registration successful	Pass

Table 4.2: Admin Modules Testing

S.N.	Test Case	Input	Expected Result	Actual Result	Status
1	Add Product	Submit valid product details	Product created	Product created	Pass
2	Edit Product	Update product attributes	Product updated	Product updated	Pass
3	Delete Product	Delete a product	Product deleted	Product deleted	Pass
4	View Order Details	Open the Orders tab	Order list displayed	Order list displayed	Pass
5	Grant Administrative Privilege	Update a user's role	Role updated	Role updated	Pass

Table 4.3: User Testing

S.N.	Test Case	Input	Expected Result	Actual Result	Status
1	Open Product Page	Select a product	Product page opens	Product page opens	Pass
2	Cart	Add items to the cart	Items added	Items added	Pass
3	Checkout	Submit valid shipping and order details	Order confirmed	Order confirmed	Pass
4	View Order History	Open profile order history	Ordered items displayed	Ordered items displayed	Pass

4.2.2 Automated Unit Testing

The forecasting and recommendation utilities were tested with deterministic inputs and expected outputs.

Table 4.4: Automated Forecasting Module Tests

S.N.	Test Case	Input	Expected Result	Actual Result	Status
1	Order Validation	Order above declared range	Reject unsupported order	Rejected with validation error	Pass
2	Seasonal Differencing	Repeating weekly series	Recover weekly forecast	Weekly pattern recovered	Pass
3	Multiplicative AR Terms	Synthetic AR series	Recover regular and seasonal terms	Terms recovered within tolerance	Pass
4	Multiplicative MA Terms	Synthetic MA series	Recover regular and seasonal terms	Terms recovered within tolerance	Pass
5	Error Measures	Known actual and forecast values	Correct MAE, RMSE, MAPE, MASE, and WAPE	Values matched	Pass
6	Nested Random Walk	Integrated synthetic series	Constant multi-step point forecast	Forecast matched last value	Pass
7	Bootstrap Intervals	Fixed random seed	Reproducible ordered 80% bounds	Bounds reproduced and ordered	Pass
8	Residual Diagnostics	Synthetic AR series	Roots and Ljung-Box results returned	Diagnostics returned	Pass
9	Chronological Split	120 ordered values	96 development and 24 test values	Split preserved order	Pass
10	Validation Origins	15-day horizon	Seven expanding origins without leakage	Origins valid for the selected horizon	Pass
11	Final-Test Origins	Protected test sequence	Cover each test value once in order	Coverage and order verified	Pass

Table 4.4: Automated Forecasting Module Tests (Continued)

S.N.	Test Case	Input	Expected Result	Actual Result	Status
12	Explicit Baselines	Synthetic sequence	Naive and seasonal-naive forecasts	Expected values returned	Pass
13	Candidate Grid	Declared order generator	72 unique SARIMA orders	72 unique orders returned	Pass
14	Training History Policy	200 ordered values with a 91-day window	Fit on the latest 91 values without changing metric history	Training sample verified	Pass
15	Pooled RMSE	Fold RMSE values with unequal test sizes	Root of the weighted squared errors	Pooled value matched	Pass
16	Coefficient Names	Fitted MA model	Standard intercept and MA coefficient names	Names and finite values verified	Pass

Table 4.5: Automated Recommendation Scoring Tests

S.N.	Test Case	Input	Expected Result	Actual Result	Status
1	Specification Parsing	Colon-delimited product description	Normalized key-value features	Features parsed correctly	Pass
2	Feature Aggregation	Repeated and distinct attribute values	Unique values retained by attribute	Values aggregated correctly	Pass
3	Feature Match Levels	Exact, partial, and absent values	Scores 1.0, 0.5, and 0.0	Expected scores returned	Pass
4	Traditional Similarity	Fully matching product and profile	Weighted score equals 1.0	Score equals 1.0	Pass
5	Description Similarity	One exact and one partial comparable feature	Mean score equals 0.75	Score equals 0.75	Pass

4.2.3 System Integration Testing

These cases verify communication among the frontend, Express backend, database operations, and forecasting artifact endpoints.

Table 4.6: System Integration Testing

S.N.	Test Case	Input	Expected Result	Actual Result	Status
1	Login and Product Page Access	User logs in and opens product page	Product page loads successfully	Product page loads successfully	Pass
2	Product Detail and Recommendations	Open a product page	Details and recommendations displayed	Details and recommendations displayed	Pass
3	Product Addition and Visibility	Administrator adds a product	Product visible in catalogue	Product visible in catalogue	Pass
4	Product Deletion and Visibility	Administrator deletes a product	Product removed from catalogue	Product removed from catalogue	Pass
5	Fifteen-Day Forecast Integration	Request period 15days	Forecast, intervals, and category summaries returned	Complete artifact returned	Pass
6	Forecast Metrics Integration	Request matching period metrics	Baselines, diagnostics, and interval coverage returned	Period-specific metrics returned	Pass

4.3 Result Analysis

The forecasting results presented in this section are generated directly from the evaluation pipeline. MAE, MAPE, WAPE, MASE, and RMSSE are aggregated across the rolling origins, while RMSE is pooled from the squared errors across all evaluated observations. Raw daily revenue is used as the primary forecasting target. Results for the trailing 3-day mean are included only to examine the effect of using a smoothed target.

Table 4.7: Forecast Evaluation Design

Component	Configuration
Primary target	Raw daily revenue (366 observations)
Development/final test	293 / 73 chronological observations
Validation	7 expanding origins for the 15-day horizon
Candidate grid	72 SARIMA orders combined with 13-week, 18-week, 26-week, and expanding histories (288 configurations)
Selection rule	Require residual adequacy and weekly seasonal dynamics, then rank by MASE, pooled RMSE, and fewer fitted terms on development origins
Baselines	Naive, seasonal naive, and selected non-seasonal ARIMA
Independent reference	Same-order statsmodels SARIMA
Aggregation	Mean metrics across rolling origins; pooled RMSE across rolling origins

Table 4.8: Stationarity Tests on the Selected Training History

Transformation	ADF p-value	KPSS p-value
Log level	0.0000	0.1000
Regular difference	0.0000	0.1000
Weekly difference	0.0000	0.1000
Regular and weekly difference	0.0010	0.1000

Table 4.9: Development Model Comparison for the 15-Day Horizon

Model	MAE	RMSE	MAPE	MASE
From-scratch selected order	23,287.27	30,666.00	16.04%	0.817
Naive	23,513.06	29,613.83	16.60%	0.824
Seasonal naive	29,816.46	38,639.65	20.69%	1.045
Non-seasonal ARIMA	24,819.54	30,777.97	17.60%	0.871
statsmodels SARIMA reference	24,261.44	30,898.00	16.32%	0.851

Table 4.10: Development Point-Forecast Method Comparison

Point method	MAE	RMSE	MAPE	MASE
Conditional log forecast	23,287.27	30,666.00	16.04%	0.817
Bootstrap mean on original scale	24,403.28	30,918.31	17.64%	0.856

Table 4.11: Selected-Model Coefficient Comparison

Parameter	Custom CSS	statsmodels MLE	Absolute difference
ar.L1	0.0665	0.0682	0.0017
ar.L2	0.0702	0.0649	0.0053
ma.S.L7	-0.7388	-0.8477	0.1089

Table 4.12: Top Forecast-Error SARIMA Candidates for the 15-Day Horizon

Order	History	RMSE	MAPE	MASE	Adequate
(2,0,0)(0,1,1) ₇	126 days	30,666.00	16.04%	0.817	Yes
(0,0,0)(0,0,0) ₇	91 days	29,587.27	15.88%	0.819	Yes
(1,0,0)(0,0,0) ₇	91 days	29,590.09	15.94%	0.821	Yes
(0,0,1)(0,0,0) ₇	91 days	29,613.43	15.96%	0.822	Yes
(0,1,0)(0,0,0) ₇	91 days	29,613.83	16.60%	0.824	No

Table 4.13: Final-Test Model Comparison for the 15-Day Horizon

Model	MAE	RMSE	MAPE	MASE
From-scratch selected order	22,944.47	28,621.42	16.97%	0.761
Naive	28,606.89	36,958.98	20.28%	0.950
Seasonal naive	30,326.88	37,363.56	22.19%	1.005
Non-seasonal ARIMA	24,021.93	28,776.36	17.82%	0.797
statsmodels SARIMA reference	22,267.89	27,412.39	16.52%	0.739

Table 4.14: Raw and Three-Day Smoothed Target Ablation

Horizon	Target	MAE	MAPE	MASE
15 days	Raw revenue	22,944.47	16.97%	0.761
15 days	Trailing 3-day mean	11,478.05	8.18%	0.625

Table 4.15: Category Quantity Results for the 15-Day Horizon

Category	Selected Order	Final MASE	Zero Days
Headphones	$(0, 1, 1)(1, 0, 0)_7$	0.997	99
Laptop	$(0, 1, 2)(0, 0, 0)_7$	0.691	2
Smartphone	$(1, 1, 1)(0, 0, 0)_7$	0.698	0
Smartwatch	$(0, 1, 2)(0, 1, 1)_7$	0.900	2
Tablet	$(0, 1, 2)(0, 0, 1)_7$	0.798	2

Table 4.16: Development-Fit Residual Diagnostics

Horizon	p (7)	p (14)	p (21)	Min. AR Root	Min. MA Root
15 days	0.2310	0.1944	0.1856	3.329	1.044

Table 4.17: Rolling Final-Test 80% Prediction-Interval Coverage

Horizon	80% Coverage	80% Mean Width
15 days	84.9%	\$85,171.38

For the 15-day horizon, 260 of the 288 order-and-history configurations converged, and 96 passed the development residual and root checks. The best adequate candidate was SARIMA(2,0,0)(0,1,1)₇ using the most recent 126 days. It achieved a validation MAPE of 16.04% and a validation MASE of 0.817. The naive validation MASE was 0.824. The selected model therefore provided a small improvement over the naive baseline while retaining a recurring weekly forecast pattern.

On the rolling final test, the selected model achieved a MAPE of 16.97% and a MASE of 0.761. The naive baseline achieved a MAPE of 20.28% and a MASE of 0.950, while the seasonal-naive baseline achieved 22.19% and 1.005, respectively. The selected model reduced final-test MAPE by 3.31 percentage points relative to the naive baseline and by 5.22 percentage points relative to the seasonal-naive baseline. It also performed better than the independently selected non-seasonal ARIMA baseline on the final period. The same-order statsmodels reference achieved a MAPE of 16.52% and a MASE of 0.739. These final-test results were not used for model selection.

The conditional log forecast and bootstrap-mean point forecast were compared on the development origins. The conditional forecast achieved a MAPE of 16.04% and a MASE of 0.817, compared with 17.64% and 0.856 for the bootstrap mean. The conditional forecast was therefore retained as the point estimator. The bootstrap-mean result is included as a sensitivity check. The point forecast is not required to equal the midpoint of the bootstrap prediction interval.

The stationarity tests in Table 4.8 use the selected 126-day development history. The log-level series had an ADF $p < 0.001$ and a KPSS $p = 0.1000$. The seasonal difference was selected based on rolling forecast performance rather than the stationarity tests alone. It preserves recent weekly changes across the 15-day forecast horizon, while the seasonal moving-average term accounts for recent weekly errors.

The selected model had Ljung-Box p -values of 0.2310, 0.1944, and 0.1856 at lags 7, 14, and 21, respectively. No significant residual autocorrelation was detected at the 0.05 level. The minimum AR and MA root moduli were 3.329 and 1.044, respectively, satisfying the stability and invertibility checks. The 80% prediction interval achieved 84.9% empirical coverage, which is close to its nominal level.

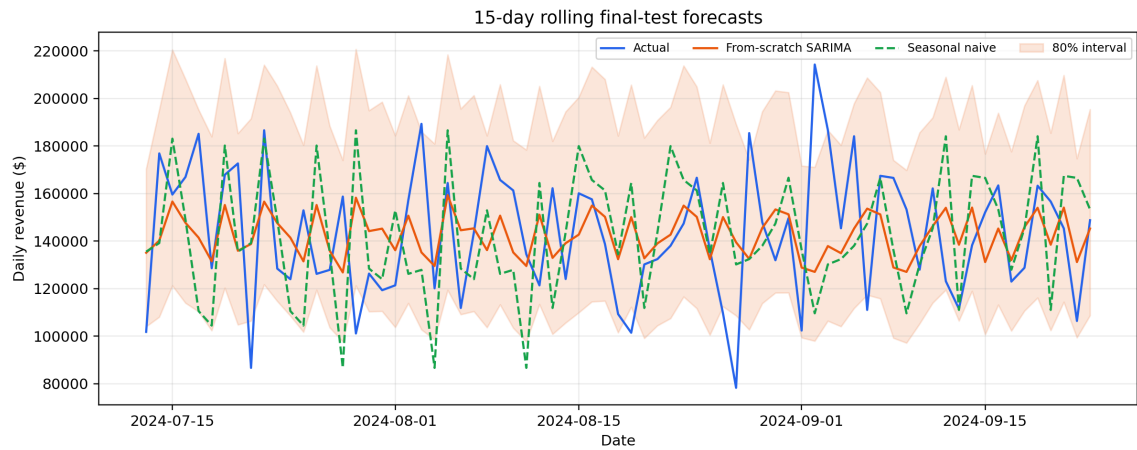


Figure 4.1: Rolling Final-Test Forecasts and 80% Prediction Intervals

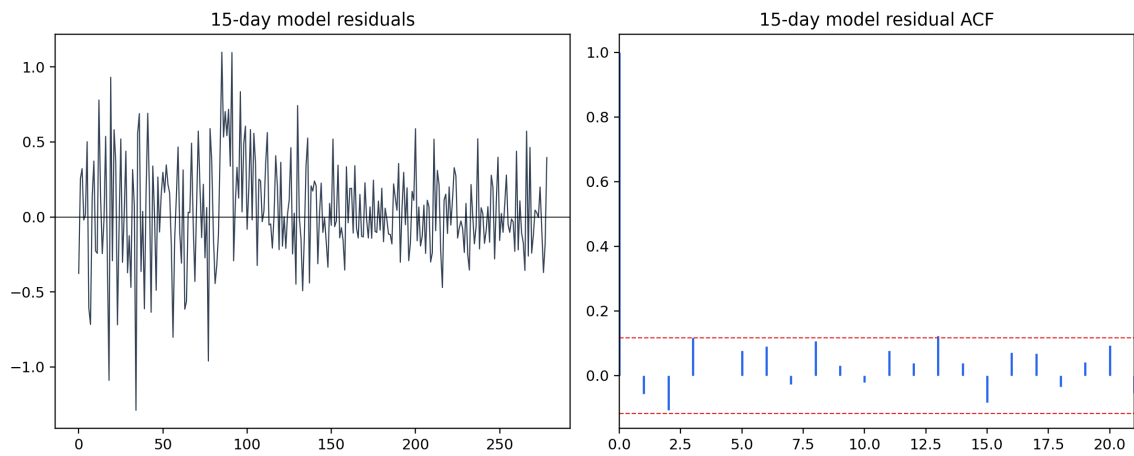


Figure 4.2: Development-Fit Residual and Autocorrelation Diagnostics

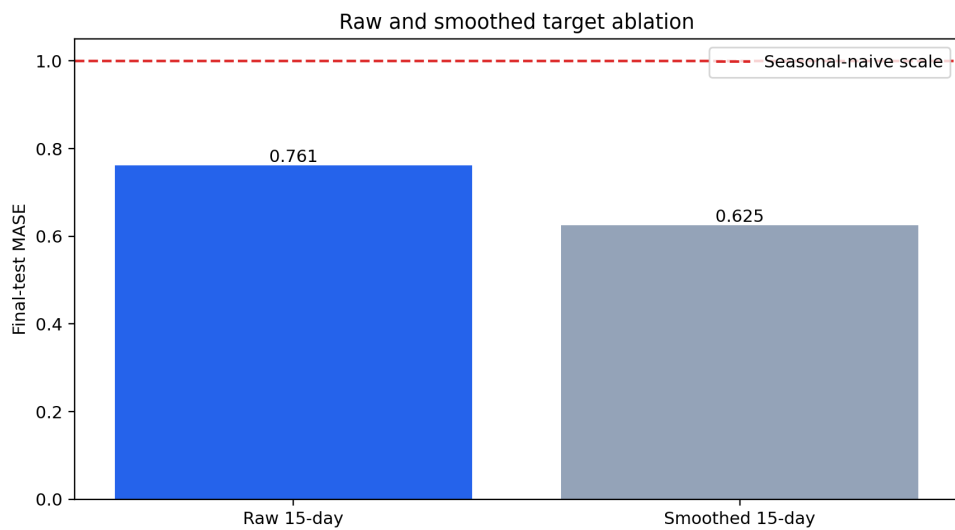


Figure 4.3: Raw and Smoothed Target Ablation

The smoothed-target errors are lower because they are calculated on a less variable series, not because the model necessarily predicts raw daily revenue better. Category results use raw daily quantity. The Headphones series contains 99 zero-sales days, so MASE is used for category order selection rather than MAPE. Category revenue shown on the dashboard is estimated by multiplying forecast quantity by the recent average category price. After evaluation, the revenue model is fitted to the most recent 126 days, while each category model is fitted to its complete available series. The resulting forecasts and 2,000-path prediction intervals are saved as versioned artifacts for the dashboard.

Content-Based Recommendation:

The recommendation module builds a user profile from deduplicated viewed, cart, and purchased product identifiers. Traditional similarity considers category, brand, price, and rating. Description similarity uses exact or substring matches between colon-delimited key-value specifications. The two component scores are combined as:

$$\text{Final Score} = 0.40 \times \text{Traditional Similarity} + 0.60 \times \text{Description Similarity}$$

Automated tests verify feature parsing, feature aggregation, exact and partial matching, and the declared weights. The returned score represents algorithmic similarity rather than recommendation accuracy. The current study does not include relevance labels, precision, recall, NDCG, or a user study. Therefore, a high similarity score is not interpreted as evidence of user preference. General recommendations rank all products according to the final score, while the product page excludes the currently viewed product from the recommendation results.

CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Conclusion

This project integrates a web-based electronics marketplace with deterministic content-based recommendation and sales forecasting. The from-scratch SARIMA framework implements data transformation, differencing, multiplicative AR and MA terms, CSS estimation, recursive forecasting, residual diagnostics, and bootstrap prediction intervals. Seventy-two SARIMA orders were combined with four training-history policies and evaluated at seven rolling origins while keeping the final test period separate. Residual adequacy checks and validation results selected the 126-day SARIMA(2,0,0)(0,1,1)₇ configuration. The selected model captured a recurring weekly pattern and achieved a lower development MASE than the naive baseline. On the protected final period, it also performed better than the naive, seasonal-naive, and independently selected non-seasonal ARIMA baselines. Comparisons with the statsmodels implementation provided an additional check of the fitted coefficients and forecast results.

The forecasting results were integrated into the marketplace through saved forecast artifacts, allowing normal dashboard requests to use stored results without refitting the model. Administrators can explicitly regenerate the artifacts when the dataset or forecasting code changes. The recommendation module uses deterministic similarity scoring based on user activity and product attributes, with automated tests covering its main scoring components. However, recommendation relevance has not been evaluated using ground-truth relevance data or user studies. Future work can address this limitation by introducing offline recommendation metrics and user-based evaluation.

5.2 Future Recommendations

Future forecasting work should use several years of sales data and include external variables such as promotions, holidays, prices, and stock availability. Prediction interval calibration could also be improved, and intermittent-demand methods could be evaluated for categories with sparse sales. The custom optimizer could be compared with full maximum-likelihood estimation using a wider but predeclared SARIMA order range. For the recommendation module, future work should introduce reproducible relevance labels, precision and ranking metrics, user-based evaluation, and collaborative or hybrid recommendation methods. These improvements would allow the analytical modules to be evaluated not only as integrated system features but also in terms of their measurable effectiveness.

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APPENDIX

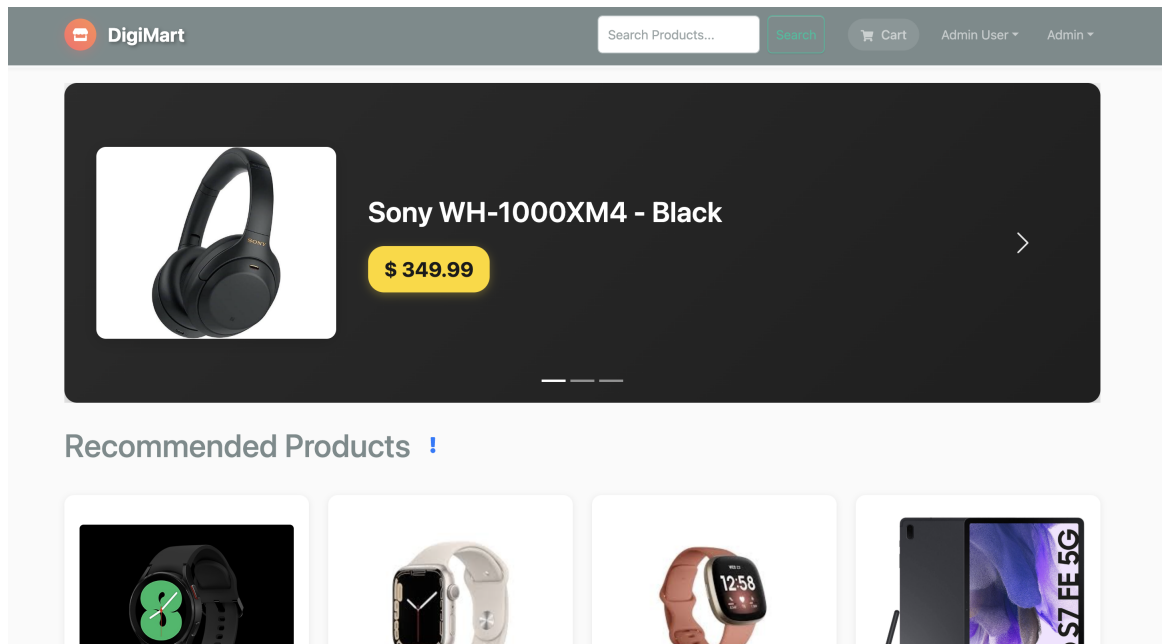


Fig 1: Home Page

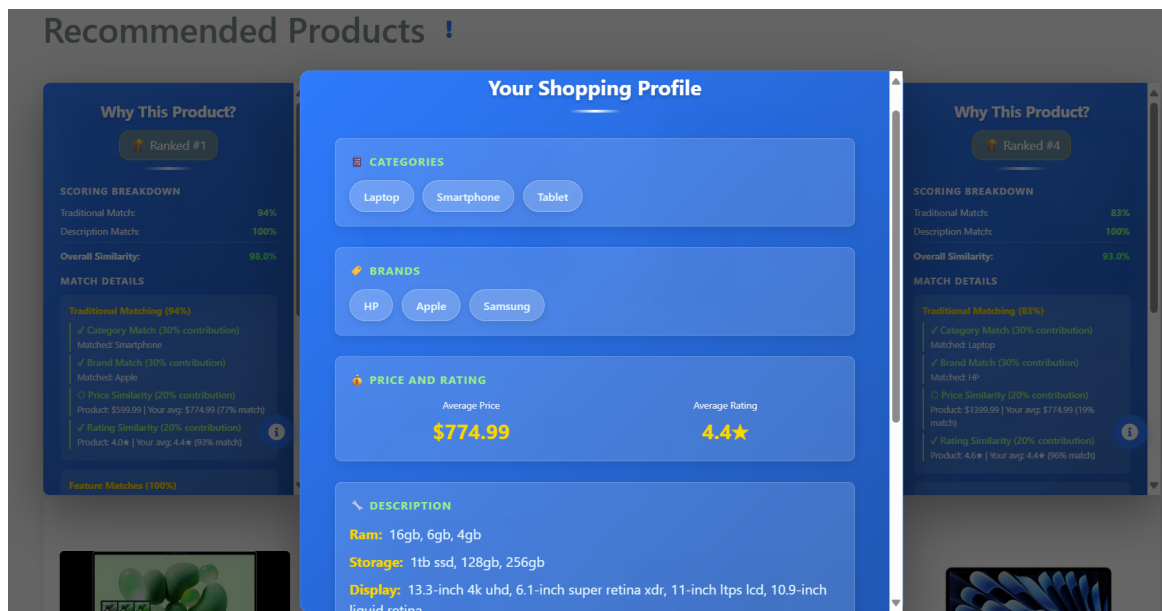


Fig 2: Recommendation Details

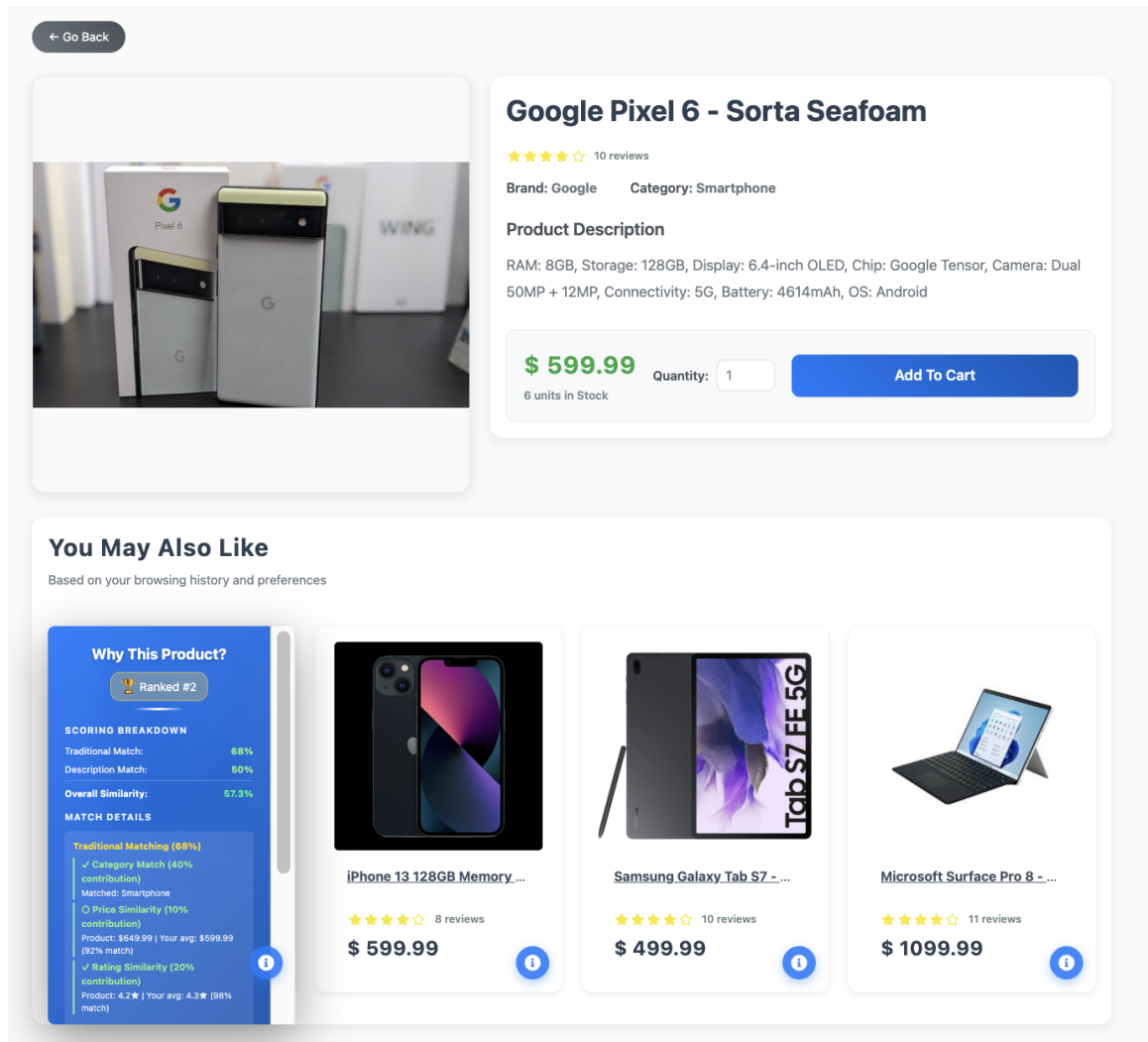


Fig 3: Product Page

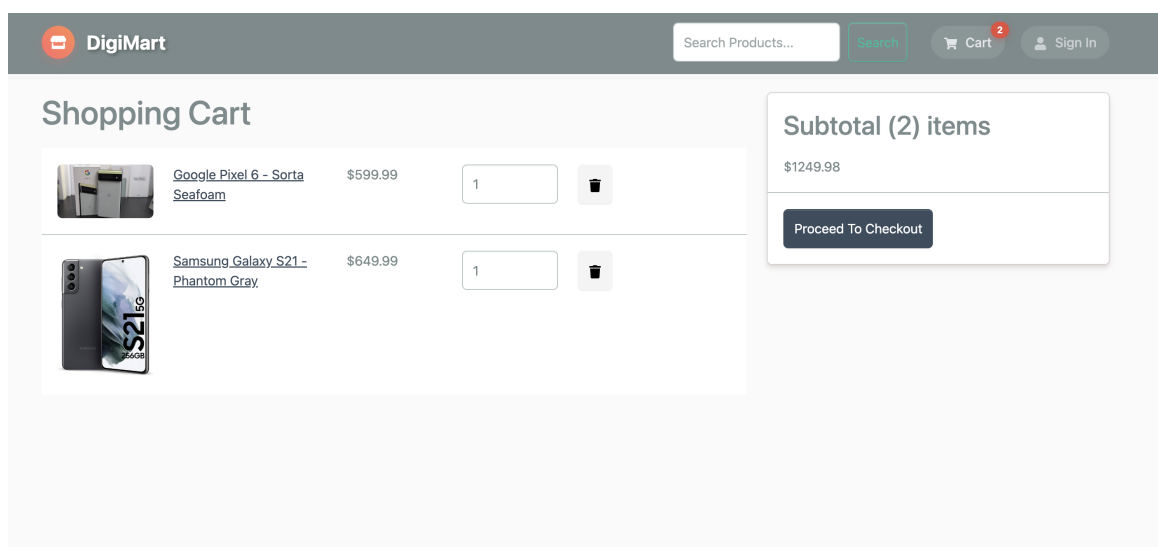


Fig 4: Cart Items

My Profile

Manage your account and view order history

Account Information

Full Name

John Doe

Email Address

ram@gmail.com

New Password

Enter new password (optional)

Leave blank to keep current password

Confirm Password

Confirm new password

Update Profile

My Orders

ORDER ID	DATE	TOTAL	ACTIONS
6a815cb05cd4120eb1b2b0e0	Aug 16, 2026	\$ 1437.48	Details

Fig 5: User Profile

Products Management

Manage your product catalog

Create Product

PRODUCT ID	NAME	\$ PRICE	CATEGORY	BRAND	ACTIONS
6a7d5f4b3da0ec0bebac507e	iPhone 13 128GB Memory - Midnight Black	\$599.99	Smartphone	Apple	
6a7d5f4b3da0ec0bebac507f	Samsung Galaxy S21 - Phantom Gray	\$649.99	Smartphone	Samsung	
6a7d5f4b3da0ec0bebac5080	Google Pixel 6 - Sorta Seafoam	\$599.99	Smartphone	Google	
6a7d5f4b3da0ec0bebac5081	Samsung Galaxy Tab S7 - Mystic Black	\$499.99	Tablet	Samsung	
6a7d5f4b3da0ec0bebac5082	Apple iPad Air - Sky Blue	\$599.99	Tablet	Apple	
6a7d5f4b3da0ec0bebac5083	Microsoft Surface Pro 8 - Platinum	\$1099.99	Tablet	Microsoft	
6a7d5f4b3da0ec0bebac5084	Dell XPS 13 Laptop - Platinum Silver	\$1199.99	Laptop	Dell	
6a7d5f4b3da0ec0bebac5085	HP Spectre x360 Laptop - Nightfall Black	\$1399.99	Laptop	HP	
6a7d5f4b3da0ec0bebac5086	MacBook Pro 14-inch - Space Gray	\$1999.99	Laptop	Apple	
6a7d5f4b3da0ec0bebac5087	Apple Watch Series 7 - Starlight	\$399.99	Smartwatch	Apple	

Fig 6: Product Management

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Admin User ▾
Admin ▾

Orders Management

View and manage customer orders

ORDER ID	CUSTOMER	DATE	TOTAL	PAYMENT METHOD	ACTIONS
6a815cb05cd4120eb1b2b0e0	John Doe	Aug 16, 2026	\$ 1437.48	COD	Details
6a815db00d949475b3654137	Ram Nepali	Aug 16, 2026	\$ 113.49	COD	Details

Fig 7: Order Management

Admin User ▾
Admin ▾

User Management

Manage user accounts and permissions

USER ID	NAME	EMAIL	ROLE	JOINED	ACTIONS
68fb7abd...64b1	Admin User	admin@gmail.com	✓ Admin	Oct 24, 2025	Protected
68fb7abd...64b2	John Doe	ram@gmail.com	✕ User	Oct 24, 2025	Edit Delete
68fb7abd...64b3	Jane Doe	niraj@gmail.com	✕ User	Oct 24, 2025	Edit Delete

Fig 8: User Management

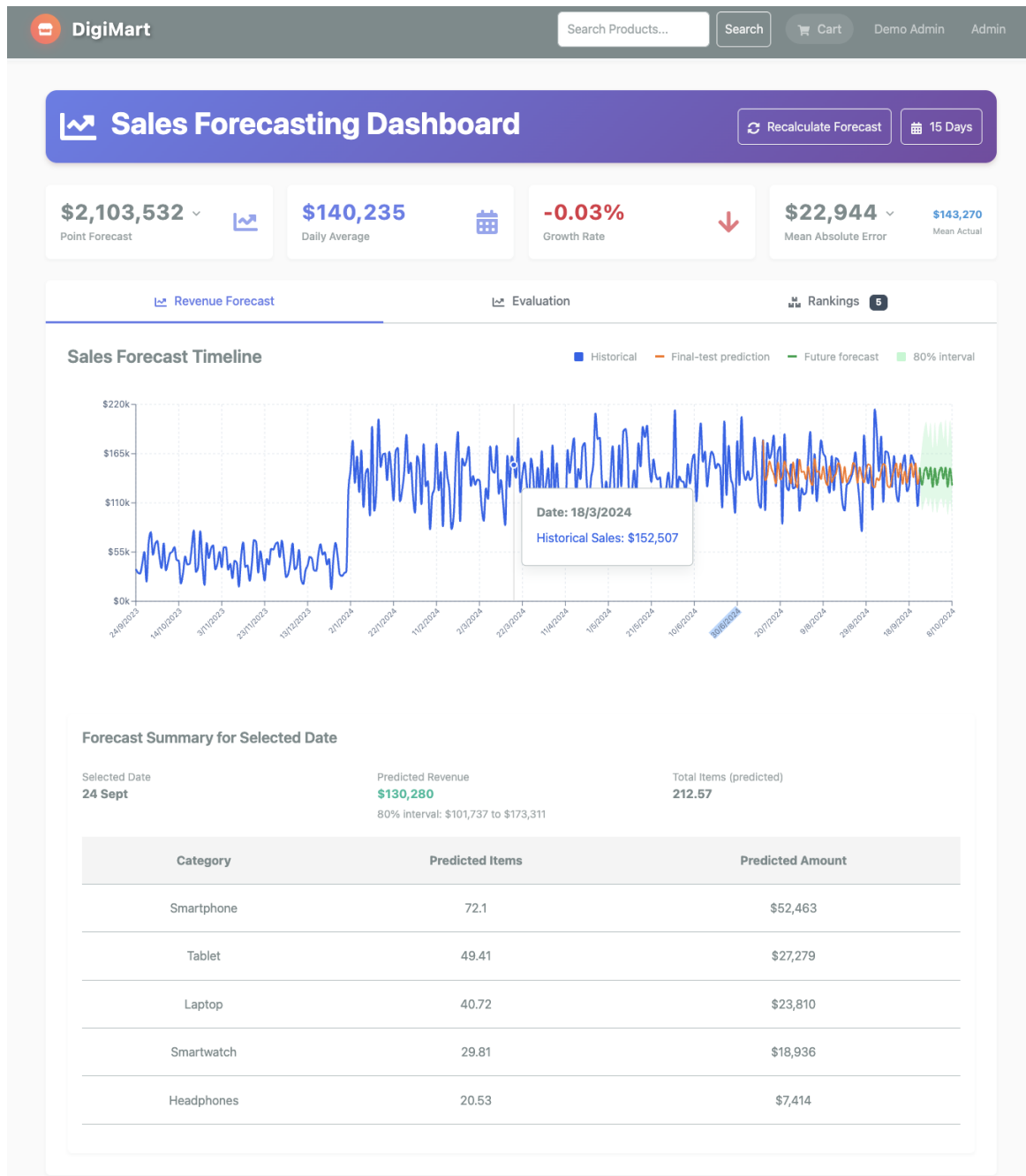


Fig 9: Sales Forecasting

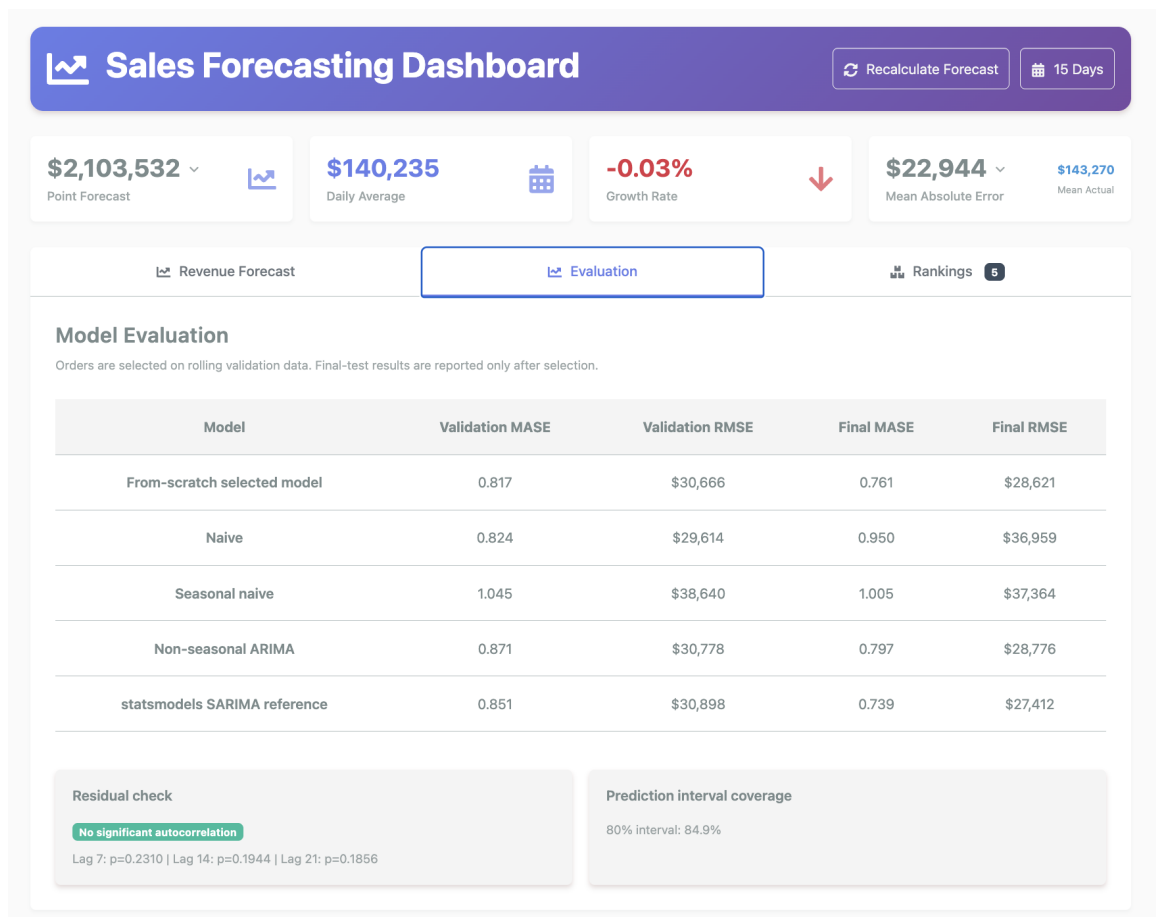


Fig 10: Model Evaluation

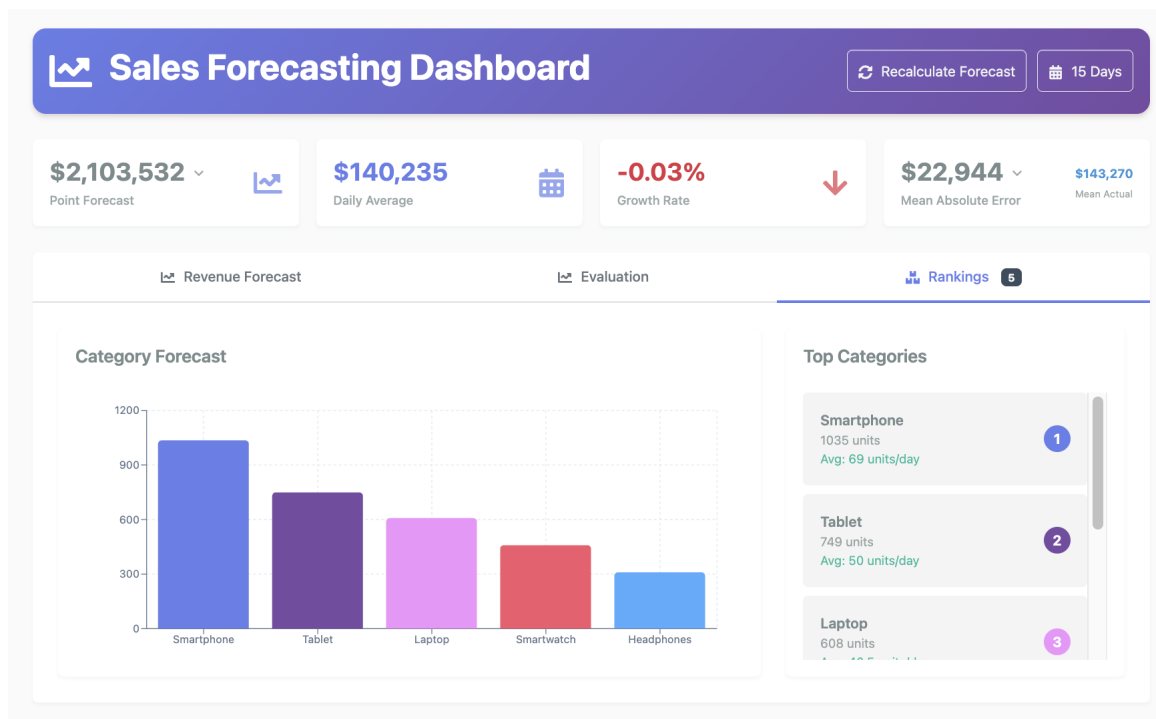


Fig 11: Forecasted Category Ranking